

Week 11

SLAM, and Merge 1

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★ Merge 1

Electrical + Chassis

The claim

The vehicle steers to a commanded angle under firmware control, wheels off the ground.

Closes bring-up **Stage 1**.

And it is the **E4 decision point**.

What must be shown

Closed-loop position tracking	$\leq 1^\circ$ steady-state, no sustained oscillation
Loop rate	≥ 200 Hz, measured
Limit behaviour	Commanded past $\pm 22.5^\circ$, clamps rather than driving into the stop
Stall detection	Hold the wheel; the stall bit sets in <code>DbwStatus.faults</code>
Freewheel on power cut	Cut power; steering goes free, not locked
R/C signal-loss timeout	Stop the Teensy emitting pulses; the motor stops

E4 fires here, or not at all

If you are at 3° and have been adjusting gains for a week, **the honest answer is to invoke E4.**

It costs you nothing — the decision record already calls it the planned action.

What **would** cost you is quietly continuing to tune while telling the standup it is nearly there.

The numbers Chassis owes everyone

Simulation and Software have both been waiting.

Merge 2 next week cannot happen without them.

For wheelbase, track width, wheel radius — show:

- the value
- **how it was measured**
- how much you trust it

SLAM

It is circular, and that is the difficulty

To build a map you need to know where you are.

To know where you are you need a map.

SLAM does both at once. It works because of **constraints**: each scan constrains the pose that could have produced it, given the map so far.

`slam_toolbox` maintains a **pose graph** — nodes are poses, edges are scan-match constraints — and re-optimises as it goes.

Loop closure is where the value is

Drive down a corridor and back.

Odometry drifted, so the return does not land on the outward journey.
The map shows the same wall **twice**, slightly apart.

Loop closure recognises a place already in the graph, adds an edge between two poses that were far apart, and re-optimises — distributing the accumulated error backwards over the whole loop.

Which is why the gate says 30 m

`slam_toolbox` closes a ≥ 30 m hallway loop; repeated observations of the same wall agree within ≤ 10 cm; the map loads in Nav2 without manual editing.

A short straight run proves nothing.

Drift has not had room to accumulate, and **no loop has closed**.

Thresholds tighter than the defaults

```
resolution: 0.05  
max_laser_range: 12.0  
minimum_travel_distance: 0.2  
minimum_travel_heading: 0.2  
scan_buffer_size: 15
```

This vehicle moves at $\leq$ walking speed: waiting 0.5 m between scans would throw away most of the data on a slow indoor run.

Inherit the defaults and the map is just **worse**,
with nothing to indicate why.

`base_link`, not `base_footprint`

B-MROVER's SLAM used `base_footprint` while its EKF used `base_link`.

MRider standardises on `base_link` **everywhere**.

There is no `base_footprint` anywhere in this stack —
if you see one, something is importing an unported config.

Transform ownership

Transform	Owner
<code>map → odom</code>	<code>slam_toolbox</code> — the drift correction
<code>odom → base_link</code>	EKF — the local pose estimate
<code>base_link → *</code>	<code>robot_state_publisher</code>

`map → odom` **is not the robot's pose.** It is the correction between where dead reckoning thinks it is and where SLAM believes it is.

When loop closure fires, that transform **jumps** — and that is correct.

Reading a bad map

Symptom	Usual cause
Doubled walls , worse after turning	Bad sensor extrinsic — Lab 5
Smeared walls	Timestamp / sync problem
Drifts steadily, never corrects	Loop never closed — no revisited place
Jumps and reshapes	Loop closure fired. Correct
Open space where a wall is	Glass, or above/below the scan plane
Robot walks through walls	TF broken, or two publishers of one transform



A bad map is evidence about the pipeline, not a verdict on SLAM.

Each symptom points at a different upstream component.

One more candidate, new this week

`mitt_dimensions.yaml` was **estimated** until today.

A wrong wheelbase produces a wrong yaw rate —
and therefore a **heading-dependent map error that looks exactly like a bad extrinsic.**

Put it at the top of your list until the measured values land.

Track work

Track	
Electrical	Stage 3 — relay MUX and E-stop; every failsafe row, incl. E-stop with the laptop off
Chassis	Mast and equipment plate; tip-over margin with the plate loaded
Simulation	Feed measured geometry into the twin; quantify the disagreement
Software	Nav2 re-derived for measured geometry — both radius parameters together

Merge 2 next week: Simulation + Software.